

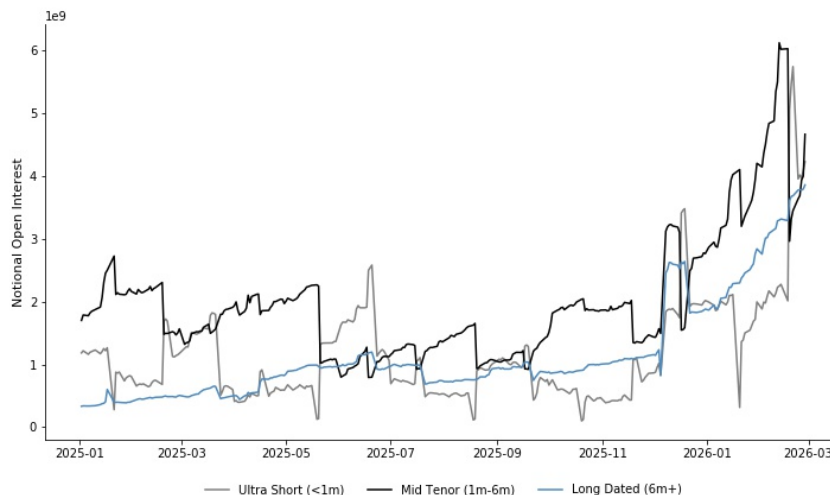
## Cross Asset Volatility

### Oil Derivatives and “Cheaper To Deliver” Macro Hedges

As geopolitical tensions evolve, we provide an update on the **oil derivatives market and trade ideas** that seek to capitalize on risk-premium dislocations. We also highlight “cheaper-to-deliver” solutions for macro hedging.

- For the oil market, we update the **flow and positioning in Crude futures and energy ETF** as well as **option positioning**.
- We identify dislocations in key parameters and propose three trades calibrated to current spot and volatility levels.
  - Short dated upside hedge - **XLE Mar 105 -115% Call spread for 0.86%**
  - “Buy Put on Rally” in oil - **Sep 95% Put on XLE KI at 110% for 2.2% or June Put Spread in Vanilla format**
  - Vol RV opportunities - **Delta Hedged Calendar in Brent Options**
- Despite macro headwinds and recent private-credit headlines, **Credit ETF and SPX** have shown muted price action at the index level. We have previously flagged the risk of [correlation increase post earnings](#) and our [approach to hedging](#). Two trades stand out as **low-premium macro hedges that benefit from elevated implied skew**:
  - **Buy SPX 6M 95% Put VKO Knock Out at 30 Vol for 1.2%**
  - **Buy HYG Apr 79-76 Put Spread for 0.42**

Figure 1: Open Interest in XLE Calls



#### Global Quantitative and Derivatives Strategy

##### Yangyang Hou <sup>AC</sup>

(44-20) 3493-1012  
yangyang.hou@jpmorgan.com  
J.P. Morgan Securities plc

##### Mengdi Wang <sup>AC</sup>

mengdi.wang@jpmchase.com  
J.P. Morgan Securities LLC

##### Ladislav Jankovic <sup>AC</sup>

(1-212) 834-9618  
ladislav.jankovic@jpmchase.com  
J.P. Morgan Securities LLC

##### Bram Kaplan, CFA <sup>AC</sup>

(1-212) 272-1215  
bram.kaplan@jpmorgan.com  
J.P. Morgan Securities LLC

##### Dobromir Tzotchev, PhD

(44-20) 7134-5331  
dobromir.tzotchev@jpmorgan.com  
J.P. Morgan Securities plc

##### Tony SK Lee

(852) 2800-8857  
tony.sk.lee@jpmorgan.com  
J.P. Morgan Securities (Asia Pacific) Limited/ J.P. Morgan Broking (Hong Kong) Limited

##### Davide Silvestrini

(44-20) 7134-4082  
davide.silvestrini@jpmorgan.com  
J.P. Morgan Securities plc

See page 12 for analyst certification and important disclosures.

J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.

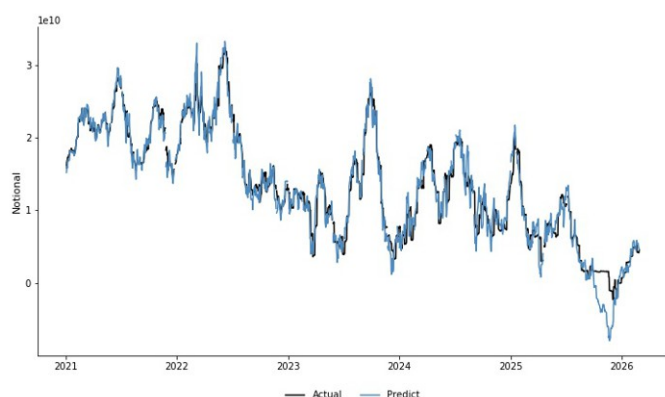
## Oil Derivatives Positioning and Trade

### Delta One Flows - Oil Futures

Oil-futures positioning declined from mid-last year, driven by Managed Money/CTA-style flows, before bottoming and rebounding in January. Commercials have increased hedging since January accordingly. [We provide daily predictions of the CFTC positions](#). A similar pattern is evident across energy ETF.

Figure 2: Managed Money for Crude Futures Notional Value

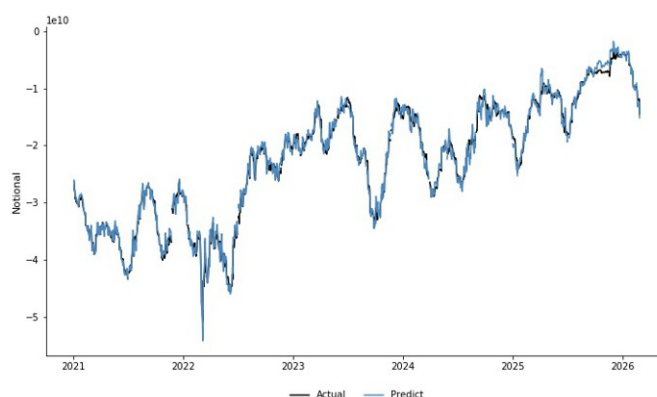
From 2021



Source: J.P. Morgan Equity Derivatives Strategy, CFTC.

Figure 3: Commercial for Crude Futures Notional Value

From 2021



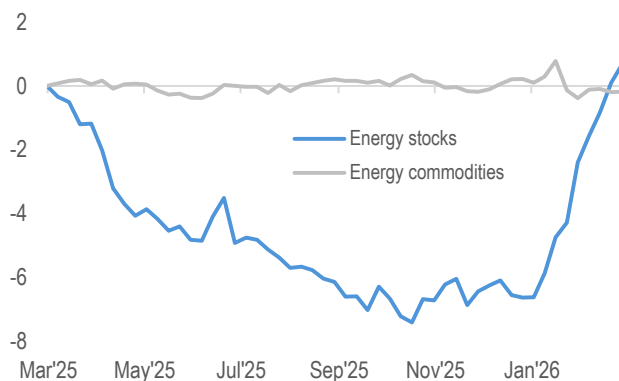
Source: J.P. Morgan Equity Derivatives Strategy, CFTC.

### Delta One Flows - Energy ETF

The energy sector was broadly out of favor through 2024-25, with persistent ETF outflows as oil prices stayed under pressure. This year, rising geopolitical risk has rekindled interest in the sector, first following the US incursion into Venezuela in January, and later with the military buildup around Iran. ETF investors have allocated significant capital to energy equities YTD, but interest in energy commodity funds has remained limited despite the geopolitical backdrop (Figure 4). See [Delta-One Flows & Positioning](#) for ongoing updates on ETF flows.

Figure 4: Energy ETF flows

Cumulative ETF flows (US listed ETFs, \$Bn)



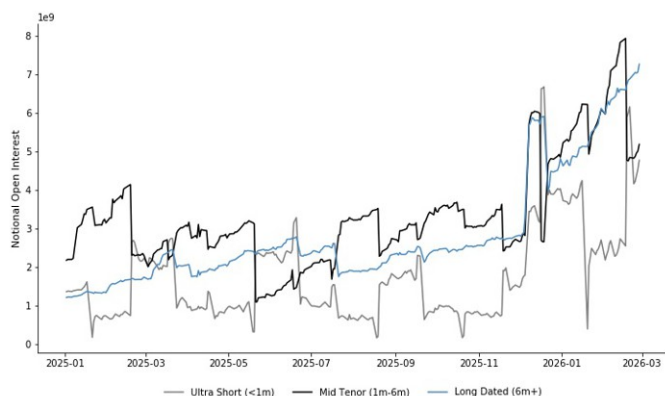
Source: J.P.Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

## Option Flows

In terms of option flows, we are able to track the energy ETF from the listed markets and a pick up in activity for both put and calls has been evident since the start of the year, with **calls increasing in volume significantly across all tenors**.

**Figure 5: Open Interest in XLE Puts**

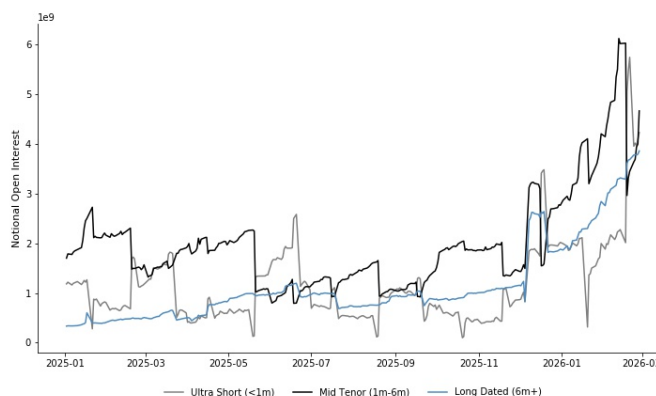
Based on Expiry Bucket in Dollar Notional



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

**Figure 6: Open Interest in XLE Calls**

Based on Expiry Bucket in Dollar Notional

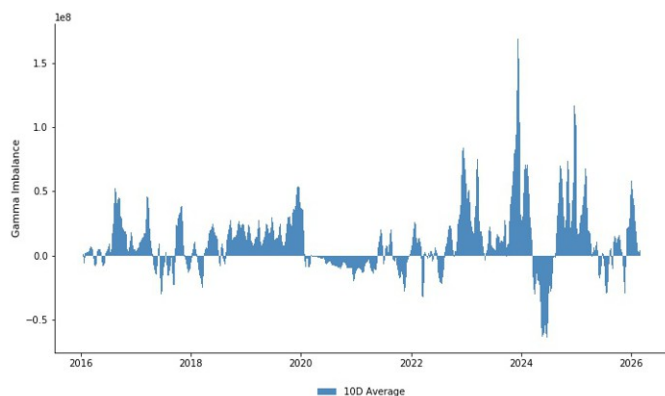


Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

As for the gamma imbalance, the order of magnitude is not as significant as what we saw in [gold options previously](#). However, dealers are **getting short gamma** from investor lifting options, **in particularly the call wing**.

**Figure 7: Average Gamma Imbalance on XLE Puts**

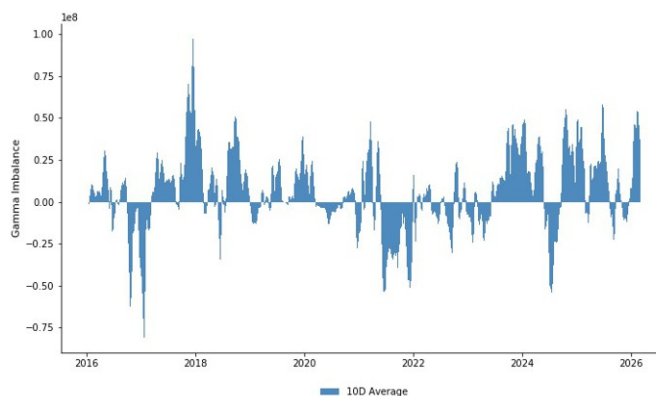
10-Day Moving Average



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

**Figure 8: Average Gamma Imbalance on XLE Calls**

10-Day Moving Average

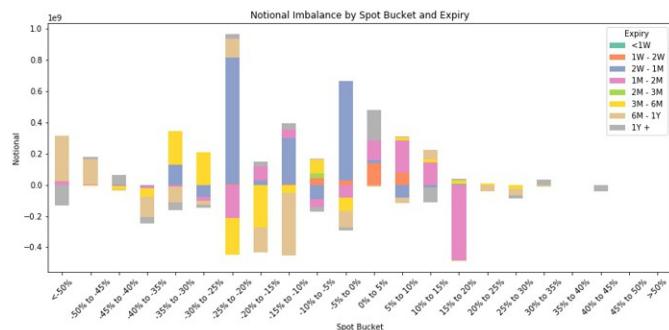


Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Current imbalance distribution indicates a more muted gamma profile on the upside, with moderate supply as a result of possible short legs of call spreads. Consequently, implied volatility increased only modestly over the weekend, though the move remains sizable relative to levels at the start of the year (Figure 13).

Figure 9: Notional Imbalance from Listed Options on XLE

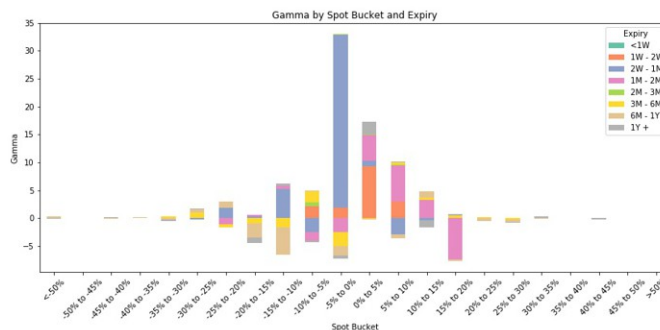
As of Feb 27<sup>th</sup>



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

Figure 10: Gamma Imbalance from Listed Options on XLE

As of Feb 27<sup>th</sup>

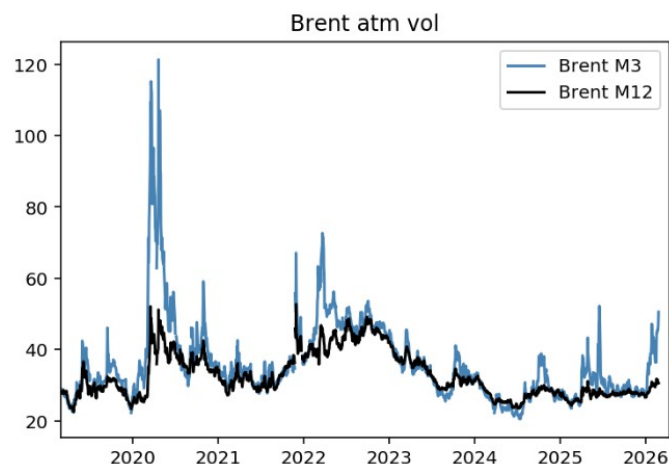


Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P. OPRA

### How Much Have Parameters Changed?

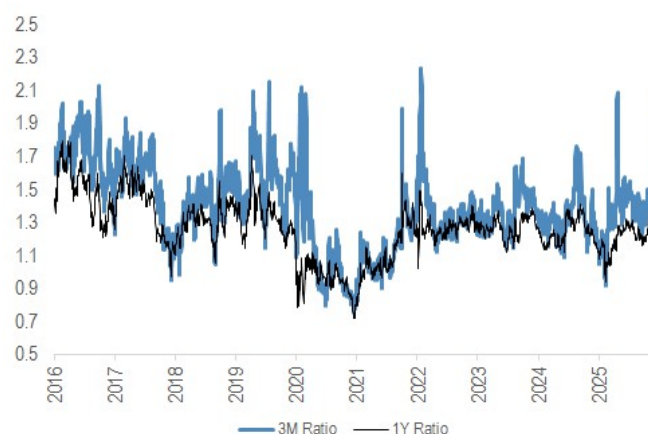
In options on oil futures, the front end of the Brent curve is reacting most acutely (**Figure 11**). M3 ATM vol surged to nearly 50% before consolidating to around 45%, still not too far from last summer's highs. This front-loaded reaction reflects heightened uncertainty, while the back end remains more anchored, with market participants expecting normalization. The Brent/XLE vol ratio is notably elevated (**Figure 12**), highlighting a **significant risk premium in crude relative to energy equities**, and suggesting that the vol market is pricing in more idiosyncratic risk for oil than is the broader energy sector.

Figure 11: Front-end vols reacted to geopolitical developments. Back end vols reaction lagging.



Source: J.P. Morgan.

Figure 12: Brent / XLE vol ratio is elevated, implying significantly more of risk premium built into Brent vols.



Source: J.P. Morgan Equity Derivatives Strategy

Figure 13: XLE Implied Vol Move from Jan 2<sup>nd</sup>

|       | 1M     | 2M     | 3M     | 6M    | 1Y    | 2Y    |
|-------|--------|--------|--------|-------|-------|-------|
| 80%F  | 4.75%  | 3.12%  | 3.08%  | 3.55% | 2.32% | 3.63% |
| 90%F  | 7.66%  | 6.38%  | 5.94%  | 4.76% | 3.41% | 4.01% |
| 100%F | 10.73% | 9.52%  | 8.24%  | 5.93% | 4.28% | 4.20% |
| 110%F | 12.04% | 10.18% | 9.70%  | 6.84% | 4.90% | 4.37% |
| 120%F | 12.13% | 9.88%  | 10.80% | 7.46% | 5.35% | 4.58% |

Source: J.P. Morgan Equity Derivatives Strategy

For XLE options, the front months have moved a significant amount compared to the trough at the beginning of the year, with **the upside tail picking up a lot more**.

## Trade Ideas

Our [commodity strategist](#) highlighted the **over supply issue** in the oil market entering into the Middle East conflict which remains in the background. That said, fluid **near-term event risks** could trigger meaningful upside price dislocations.

With greater clarity likely in the coming weeks, we favor very short-dated, **digital-like call spreads on XLE** to harness elevated **front-month upside skew** as a near-term hedge.

### Buy XLE Mar-26 60- 63 Call Spread for 0.49 Indicatively

(Ref 57.04, ~105.2%/110.5% strikes, ~0.86% premium, 6.1x Pnl leverage)

To express a view that prices will revert towards the **forward curve** after a potential near-term spike, we favor a **long-dated** upside-trigger knock-in put with **the barrier being monitored continuously**.

### Buy XLE Sep 95% Put on XLE KI at 110% for 2.2% (Vanilla Put 5.3%)

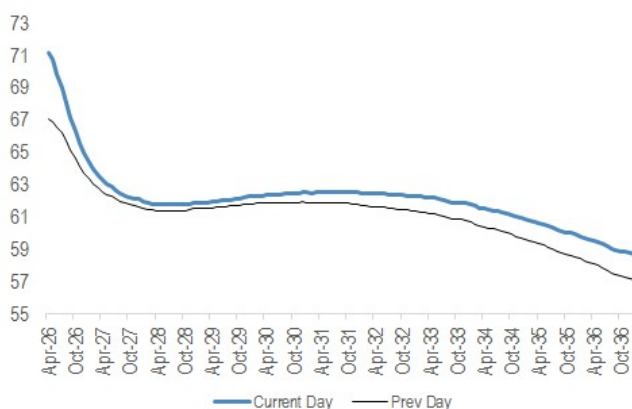
In vanilla format, we favor the June put spread for longer dated downside hedges while maintaining liquidity in bid-offers.

### Buy XLE June 55-51 Put Spread for 1.47 Indicatively

(Ref 57.04, ~96.4%/89.4% strikes, ~2.58% premium)

Figure 14: Forward Curve of Oil Futures

As of Mar 2<sup>nd</sup>



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

### Brent -M6 / +M12 calendar

The pickup in vol curve inversion has pushed fwd vols below 20—levels only seen during the COVID shock and briefly in 2025 (**Figure 17**). The dislocation between front and back vols implies that vol curve RV trades offer attractive entry points, especially when structured **gamma neutral** to avoid exposure to short-term geopolitical risk. Skew

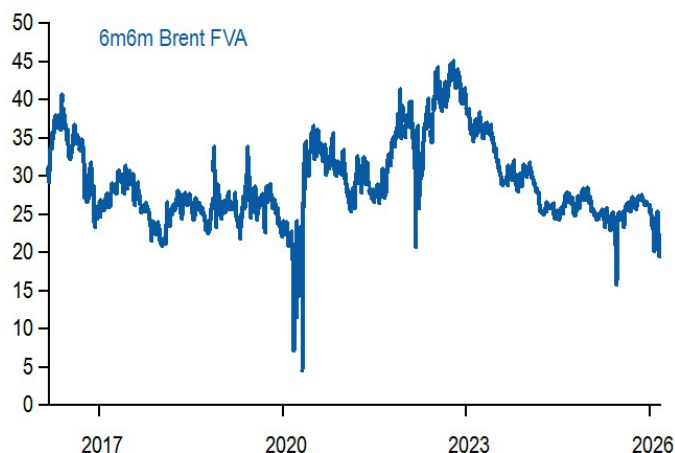
is sharply higher as well, but the firm pricing is concentrated in the front, leaving back-end skew relatively benignly priced (**Figure 18**). This allows for owning back upside vols at more reasonable prices, further enhancing the risk-reward profile of the trade.

Given these dynamics, a short M6 / long M12 Brent calendar spread, structured gamma neutral and long vega, is well supported. The trade benefits from the normalization of front end vols, and effectively hedging that via a back end skew long exposure.

Historical performance of similar constructs has been robust (**Figure 19**). Implementing the trade as a buy of M12 25 delta call versus a sell of M6 ATM call in **gamma neutral notional** captures these relative value opportunities while maintaining risks in check.

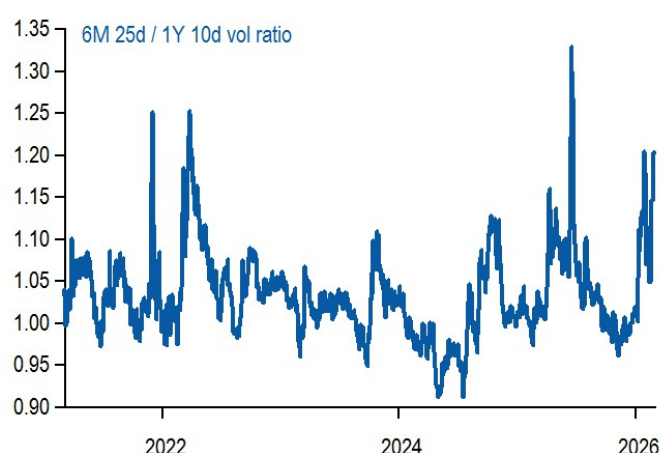
— *Buy delta-hedged M12 25 delta call @29% indic vs sell M6 atm call @36.6%, in gamma neutral notional*

Figure 15: Fwd vols dropped to below 20 vols.



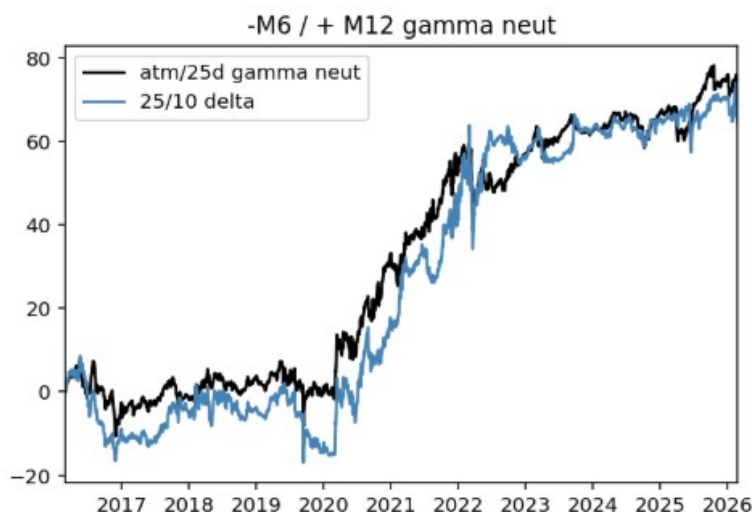
Source: J.P. Morgan Global Markets Strategy.

Figure 16: Skew popped higher, but largely so in the front.



Source: J.P. Morgan Global Markets Strategy.

Figure 17: Cumulative returns from -M6 / +M12 gamma neutral vega long calendar constructs that are also long skew have been holding well.



Source: J.P. Morgan Global Markets Strategy.

## “Cheaper To Deliver” Macro Hedges

### SPX VKO Put

SPX realized moves have remained contained due to [low correlation](#), even as **downside skew and convexity have been bid** to historically elevated levels (Figure 18). We have previously flagged the risk of [correlation increase post earnings and hence index vol levels](#).

The downside move echoes the **orderly 2022 correction** - when many volatility-linked hedges underperformed, what worked around then were simple **directional puts and VKOs**, which materially reduced protection costs by monetizing skew. We prefer longer-tenor VKOs so that realized volatility is averaged over more observations. It is worth noting that the **30v knockout barrier on 6M realized volatility was never breached even in 2022** (Figure 19).

Therefore we favor the SPX vko put with longer tenor, indicatively,

**Buy SPX 6M 95% Put 30v VKO 1.20% at offer, (vs 3.10% vanilla ref 6855.53)**

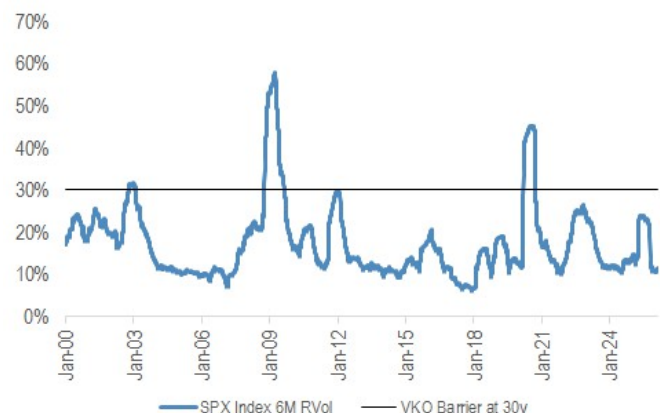
Figure 18: SPX downside skew/convexity at very elevated levels

1Y ATM - 80% Vol Spread



Source: J.P. Morgan Equity Derivatives Strategy.

Figure 19: SPX 6M Realized Vol vs VKO Barrier



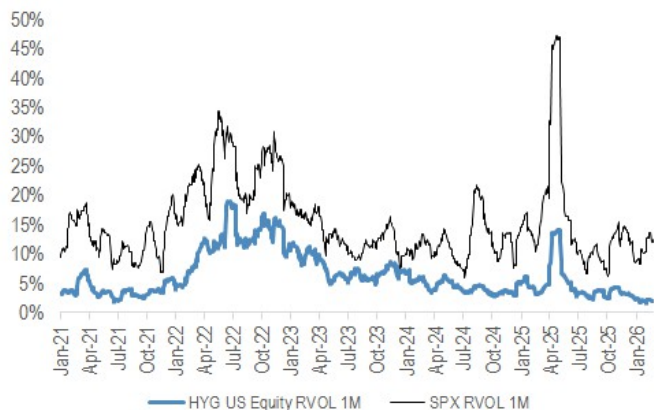
Source: J.P. Morgan Equity Derivatives Strategy.

### HYG Put Spread

While the software-led sell-off has pressured private-equity and levered-growth exposures, HYG has stayed resilient: spot remains **range-bound on the highs** and HYG **realized vol muted** versus the pickup in equities (Figure 20). Meanwhile, The SPX-to-HYG implied-volatility ratio is grinding higher (Figure 21). [JPM Credit strategists](#) also highlighted the spread widening risk and stronger demand for downside protection has **steepened HYG skew**.

Figure 20: HYG Realized Vol vs SPX Realized Vol

1M realized Vol



Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

Figure 21: SPX/HYG Implied Vol Ratio is High

2M Implied Vol

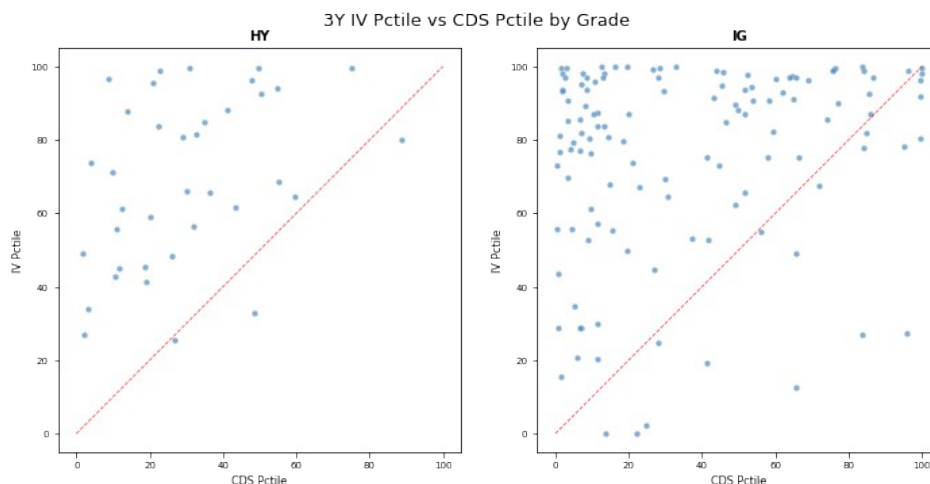


Source: J.P. Morgan Equity Derivatives Strategy, Bloomberg Finance L.P.

On the micro side, because deep out-of-the-money equity puts are economically akin to credit protection on the same issuer, we can **assess relative risk premia across equity and credit**.

Figure 22: SS Implied Vol Percentile vs CDS Percentile (3Y) by Grade

Implied Vol: 1Y 60% put, CDS: 5Y CDS



Source: J.P. Morgan Equity Derivatives Strategy.

We proxy equity protection with a 1Y 60% put, and benchmark it against 5Y CDS spreads across a universe of roughly 190 names with both actively traded options and CDS. On a **3-year percentile basis**, the observations **cluster in the upper left corner**, indicating that **equity risk premia are being priced materially higher than in CDS**.

We also show the comparison at sector level (Figure 23). The pattern is broadly consistent, indicating the **equity–credit premium gap is widespread rather than confined to a single industry**.

Against this backdrop, **HYG put spreads screen as a cost-efficient macro hedge**, leveraging the steeper skew to keep premium costs contained while targeting protection

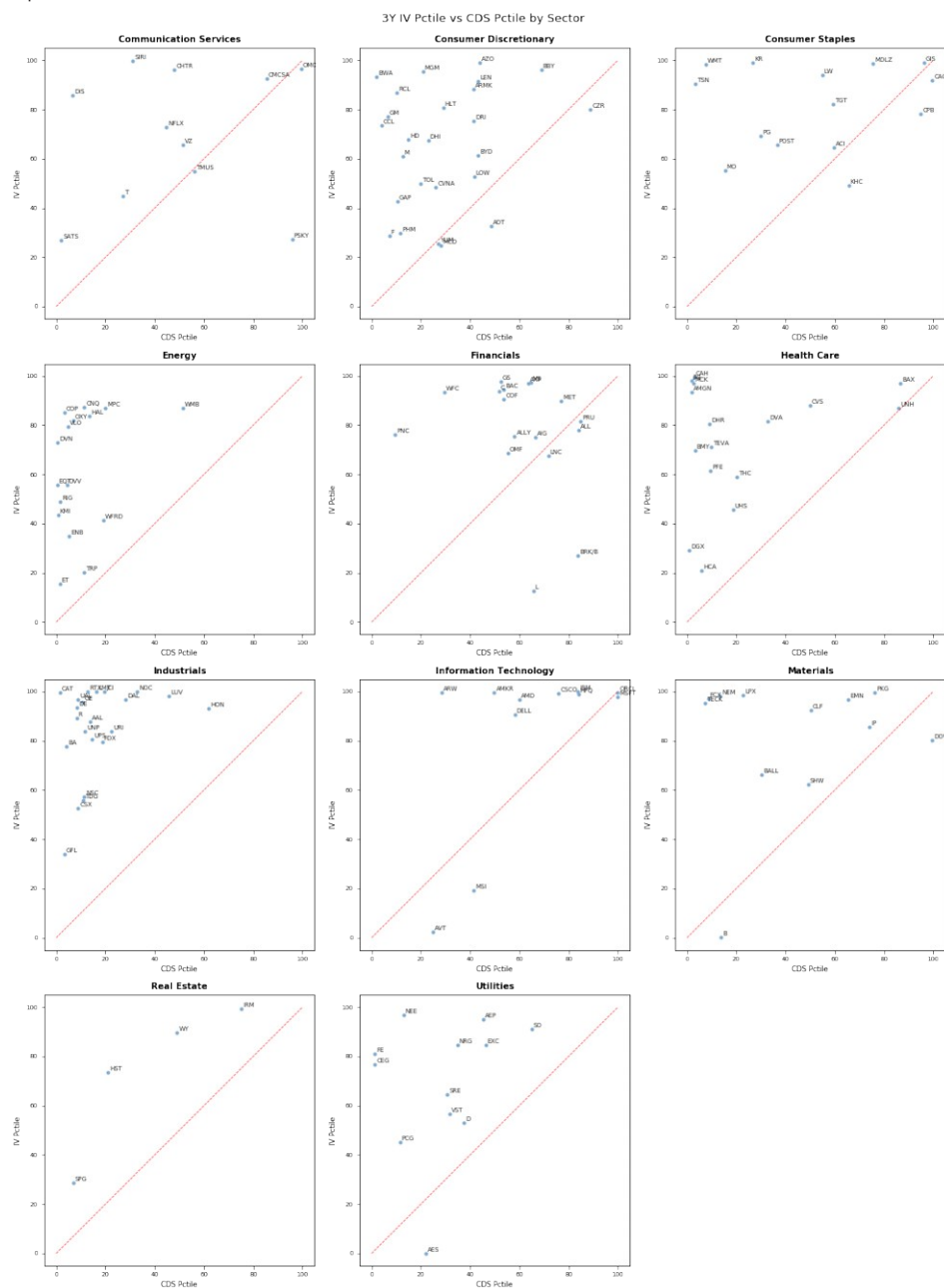
against a pullback in high yield. We reiterate our preference for HYG put spreads, consistent with [our prior recommendation](#), indicatively,

**Buy Apr-26 79-76 Put Spread on HYG for 0.42** (Ref 80.28, Delta -0.21, ~98.4%/94.5% strike for 0.52% premium, ~ 7.14 Pnl leverage)

## Equity Protection Premium vs Credit Protection Premium

Figure 23: SS Implied Vol Percentile vs CDS Percentile (3Y) by Sector

Implied Vol: 1Y 60% Put , CDS: 5Y CDS



Source: J.P. Morgan Equity Derivatives Strategy

## Appendix - Related Publications

### Cross Asset Volatility Strategy

- [Cross Asset Volatility: Gold Through the Lens of Derivatives: Positioning and Trade](#)
- [Cross Asset Volatility: Equity-USD, Buy the Dip, Japan and ML Model Update](#)
- [Cross Asset Volatility: FOMC Implied Moves, AI Credit Hedge and ML Model Output Update](#)
- [Cross Asset Volatility: Gold ETF Option flow, FTSE vs Sterling and Machine Learning Model Output Update](#)
- [Cross Asset Volatility: Record Short Gamma in Gold ETF and Machine Learning Based Trade Recommendations](#)

### Equity Volatility Strategy

- [Equity Volatility Strategy: Vol RV opportunities amid the software correction](#)
- [Equity Volatility Strategy: What to Long, Short, and When \(Part I\)](#)
- [Equity Volatility Strategy: Why index vol is so low and MXWO dispersion](#)
- [Equity Volatility Strategy: Term-structure inversion, NVDA, Gamma dispersion and VIX](#)
- [Equity Volatility Strategy: Dispersion Update, AI vs Dot-Com and Earnings](#)
- [Equity Volatility Strategy: Take some chips off the table, replace with leveraged options](#)
- [Equity Volatility Strategy Thinking Outside the Box for Tail Trading](#)
- [Equity Volatility Strategy Thinking Outside the Box for Tail Trading - Part 2](#)

### FX Derivatives Strategy

- [FX Derivatives: Dollar stall, softer vol](#), Feb 27
- [FX Derivatives: Pause and Collect](#), Feb 20
- [FXO Quant: Active, Yet Selective, Risk Appetite](#), Feb 13
- [FX Derivatives: DD: Defensive Detours](#), Feb 6

## Disclosures

**Analyst Certification:** The Research Analyst(s) denoted by an "AC" on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an "AC" on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst's personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst's compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst's analysis was made in good faith and that the views reflect the Research Analyst's own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

## Important Disclosures

**Company-Specific Disclosures:** Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan-covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing [research.disclosure.inquiries@jpmorgan.com](mailto:research.disclosure.inquiries@jpmorgan.com) with your request.

### Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, <https://www.jpmorganmarkets.com>.

### J.P. Morgan Equity Research Ratings Distribution, as of January 01, 2026

|                                              | Overweight<br>(buy) | Neutral<br>(hold) | Underweight<br>(sell) |
|----------------------------------------------|---------------------|-------------------|-----------------------|
| J.P. Morgan Global Equity Research Coverage* | 51%                 | 37%               | 12%                   |
| IB clients**                                 | 47%                 | 47%               | 35%                   |
| JPMS Equity Research Coverage*               | 49%                 | 38%               | 13%                   |
| IB clients**                                 | 72%                 | 70%               | 53%                   |

\*Please note that the percentages may not add to 100% because of rounding.

\*\*Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

**Equity Valuation and Risks:** For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email [research.disclosure.inquiries@jpmorgan.com](mailto:research.disclosure.inquiries@jpmorgan.com). For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

### History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

**Analysts' Compensation:** The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

**Registration of non-US Analysts:** Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

## Other Disclosures

---

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

**UK MIFID FICC research unbundling exemption:** UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

**Exchange-Traded Funds (ETFs):** J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

**Options and Futures related research:** If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or [https://www.finra.org/sites/default/files/2020-08/Security\\_Futures\\_Risk\\_Disclosure\\_Statement\\_2020.pdf](https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf) for a copy of the Security Futures Risk Disclosure Statement.

**Changes to Interbank Offered Rates (IBORs) and other benchmark rates:** Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: [https://www.jpmorgan.com/global/disclosures/interbank\\_offered\\_rates](https://www.jpmorgan.com/global/disclosures/interbank_offered_rates)

**Private Bank Clients:** Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

**Legal entity responsible for the production and distribution of research:** The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

## Legal Entities Disclosures and Country-/Region-Specific Disclosures:

**Argentina:** JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina (“BCRA”- Central Bank of Argentina) and Comisión Nacional de Valores (“CNV”- Argentinian Securities Commission - ALYC y AN Integral N°51).

**Australia:** J.P. Morgan Securities Australia Limited (“JPMSAL”) (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to “wholesale clients” (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia’s Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

**Brazil:** Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / [ouvidoria.jp.morgan@jpmchase.com](mailto:ouvidoria.jp.morgan@jpmchase.com).

**Canada:** J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

**Chile:** Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

**China:** J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

**Colombia:** Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC).

**Dubai International Financial Centre (DIFC):** JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

**European Economic Area (EEA):** Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE (“JPM SE”), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions (“EEA professional investors”). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

**Hong Kong:** J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

**India:** J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a ‘Research Analyst’ having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmipl.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Prasanna Bandal; [prasanna.bandal@jpmchase.com](mailto:prasanna.bandal@jpmchase.com); +912261575159. Grievance Officer: Ramprasadh K, [jpmipl.research.feedback@jpmorgan.com](mailto:jpmipl.research.feedback@jpmorgan.com); +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpmipl.com/#research>.

**Indonesia:** PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

**Korea:** J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A.,

Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

**Japan:** JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

**Malaysia:** This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

**Mexico:** J.P. Morgan Casa de Bolsa, S.A. de C.V. and J.P. Morgan Grupo Financiero are members of the Mexican Stock Exchange and are authorized to act as a broker dealer by the National Banking and Securities Exchange Commission.

**New Zealand:** This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

**Philippines:** J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

**Singapore:** This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

**South Africa:** J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

**Taiwan:** J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

**Thailand:** This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission, and its registered address is 3rd Floor, 20 North Sathorn Road, Silom, Bangrak, Bangkok 10500.

**UK:** Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

**U.S.:** J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

**General:** Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should

be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

**Confidentiality and Security Notice:** This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

**MSCI:** Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

**Sustainalytics:** Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised February 07, 2026.

---

**Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.**

Completed 03 Mar 2026 10:08 AM GMT

Disseminated 03 Mar 2026 10:08 AM GMT